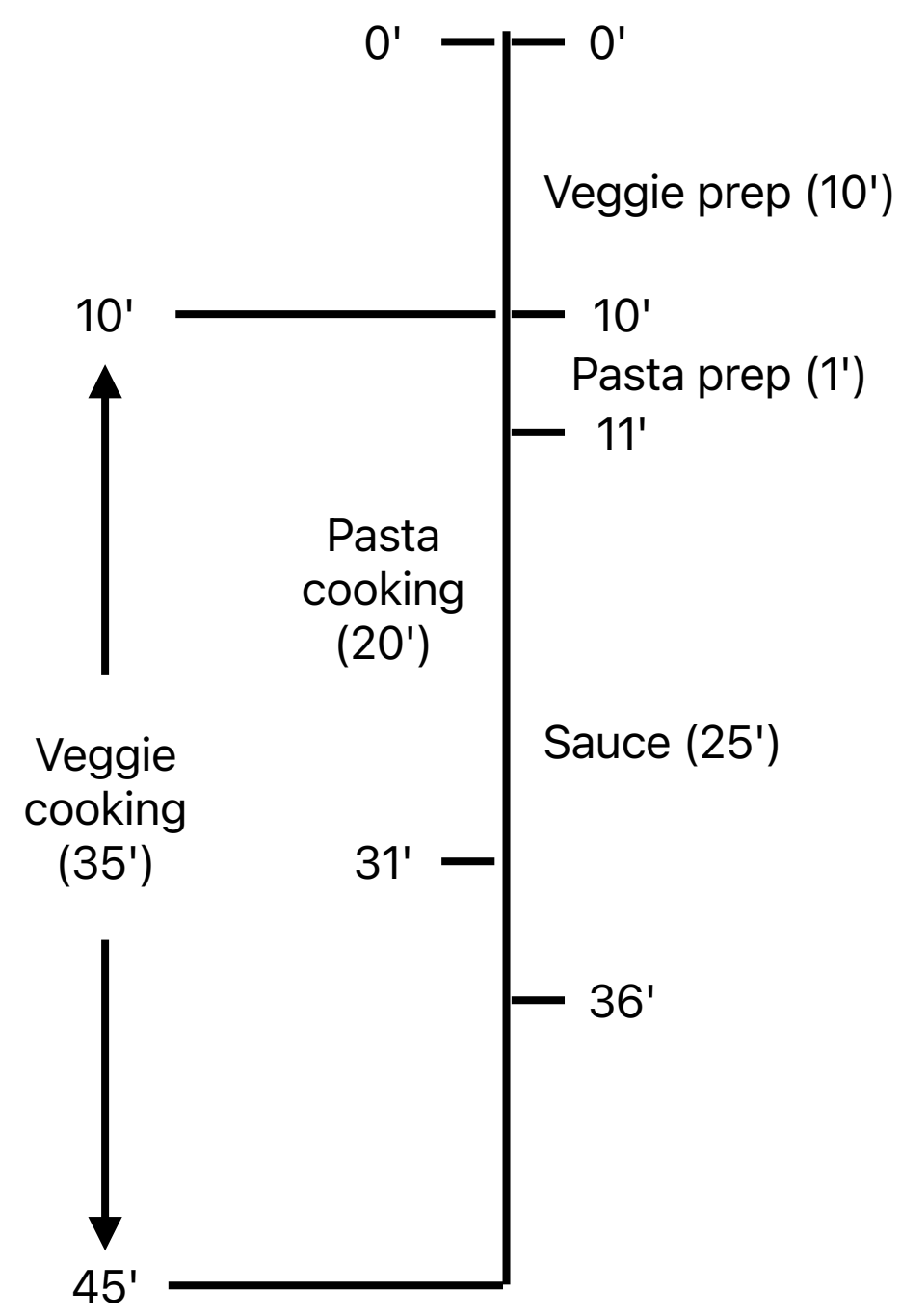


1. A cook must roast vegetables, boil pasta, and make a sauce. Preparing the vegetables and putting them in the oven takes 10 minutes, after which they roast unattended for 35 minutes. Boiling and draining the pasta takes 20 minutes and does not require the cook's attention after the first minute. Making the sauce takes 25 minutes and requires continuous attention. The vegetables can roast while the cook does other work.

What order should the cook use to minimize the total time until all three parts of the meal are ready? Draw the process diagram. How long does the meal take to make?



2. A cook must prepare roast chicken, roasted potatoes, steamed vegetables, and gravy. Preparing and seasoning the chicken takes 15 minutes of active work, followed by 60 minutes in the oven unattended. Preparing the potatoes takes 20 minutes of active work, followed by 35 minutes in the oven unattended. The oven can hold both the chicken and potatoes at the same time. Preparing and steaming the vegetables takes 18 minutes of continuous attention. Making the gravy takes 12 minutes of continuous attention, but it cannot begin until the chicken has finished because the gravy uses juices from the roasting pan.

The cook wants all four items finished as early as possible. In what order should the cook perform the active tasks, and what is the minimum total elapsed time? Draw a process diagram for this order.

